INPUT

Customer
touch screen, coins, the helmet put in the box

Box input devices

- **Coin acceptor**
  coin pulses: the main controller books every accepted coin
- **Sensors**
  helmet (ultrasonic) and the doors: enclosure, panel, back, side

Owner / Technician
phone or laptop on Wi-Fi JollyHelmet-Local: settings, rules, maintenance, backup

Team (ENTRY board)
types the previous customers (history.json)

PROCESS

JOLLY HELMET BOX SYSTEM
Box 1 · Box 2 · Box 3 main controllers (ESP32-S3)
the touch display shows the steps; the coins give the time (coin rate in Settings): welcome, coins, mode, cleaning, drying

Cleaning
UV tubes + mist nozzles

Drying
heater + fan (auto-dry at the end)

Refilling Station
ESP32-WROOM: 3 water-level sensors, 3 valves, one valve at a time, EMPTY first, then LOW

levels (radio)

ESP-NOW radio, channel 1

Local Server (Offline)
ESP32-S3, Wi-Fi JollyHelmet-Local: Control Center, session log, totals, settings

totals, sessions every 10 s

Online Server
ESP32-S3, listens only (boxes, Refilling Station, Local Server): snapshot, statistics, alerts

HTTPS upload every 1 min

Blynk Cloud
the online database: private V0 (live data), public V0 (box steps), V5 (nicknames)

HTTPS GET every 10 s / 30 s

Website
on Vercel: Monitor, Alerts, Reports, customer page. The browser reads Blynk itself

Local Database
settings (flash), session log + totals (FFat)

Backup and history
CSV log, history.json, day viewer, restore

settings, rules, fixes

OUTPUT

Box outputs
Relay controller (UV, mist, heater, fan, door locks) · TFT display + buzzer · Refilling Station valves

- **Cleaned and dried helmet**
  UV and mist cleaning, then heater and fan drying
- **Box screen and beeps**
  step, time left, coins; the OUT OF ORDER screen
- **Main container refilled**
  level EMPTY, LOW, MID, HIGH or MAX

session log and Control Center pages

- **Session record**
  date, time, box, Finished or Cancelled, coins, minutes
- **Control Center**
  Dashboard, Box 1-3 pages, Refilling Station page, Settings
- **Online Monitor**
  boxes drawn as machines, level dials, network map
- **Alerts**
  coloured by priority, sounds, push to the Blynk app
- **Reports**
  revenue and customers per day, busiest hours, what each box earns
- **Customer status page**
  FREE / IN USE / OUT OF ORDER per box, from the QR sticker

FEEDBACK AND IMPROVEMENT

- **Alerts and faults**
  box OUT OF ORDER, board not active, door opened during a run, container low
- **Refill problems**
  no level rise in time = FAILED, retry every 60 s; station silent for 15 s = box offers drying only
- **Reports and rules**
  revenue, customers and busiest hours: the owner changes the coin rate and the most coins
- **Crash reports and updates**
  last 3 crash reports in Settings, then fixes and firmware updates (re-flash)